

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) and adhering to the "Coding Requirements" section:

FR-PS-001: Generate Source Record Count Output

Title

Generate Source Record Count Output for Reconciliation

Description

The system must produce a control output containing the source record count extracted from the staging source dataset for reconciliation purposes.

Preconditions

- The Oracle staging source `STG\_E2E\_AC\_TXDBH\_DATA` is available.
- The parameter `\$\$M\_SRC\_SQL` is defined and contains valid SQL extraction logic.

Main Flow / Functional Steps

1. Read Records from Oracle Staging Source

- The system reads records from `STG\_E2E\_AC\_TXDBH\_DATA` using the SQL defined by `\$\$M\_SRC\_SQL`.

2. Apply Source Extraction SQL

- The SQL query defined in `\$\$M\_SRC\_SQL` is applied to filter and extract the required dataset from `STG\_E2E\_AC\_TXDBH\_DATA`.

3. Aggregate to Derive Source Record Count

- The system aggregates the extracted dataset to derive a single value: `SOURCE\_REC\_COUNT`.
- The aggregation logic must be one of the following:
- `SOURCE\_REC\_COUNT = COUNT()` for the total row count.
- `SOURCE\_REC\_COUNT = COUNT(T\_TX\_REQUEST\_REQUEST\_STATUS)` for counting non-null request statuses.

4. Write Result to Flat File Target

- The derived `SOURCE\_REC\_COUNT` is written to a flat file target named `FF\_Source\_Count` as `SOURCE\_RECT`.
- The output must be a single record per run.

5. Data Type Conversion and Compatibility

- Ensure that the data type of `SOURCE\_REC\_COUNT` is compatible with `SOURCE\_RECT` (string, precision 10).
- If `SOURCE\_REC\_COUNT` is numeric, apply an explicit conversion to string, ensuring it fits within the length of 10 characters.

Additional Requirements

- Handling of Null Request Status: If `COUNT(T\_TX\_REQUEST\_REQUEST\_STATUS)` is used, rows with null `T\_TX\_REQUEST\_REQUEST\_STATUS` are excluded from the count.
- Output Formatting: The count output must not exceed 10 characters when rendered as text. If counts may exceed this, either increase the target precision or implement overflow handling per operational standards.
- Reconciliation Control Output: This mapping serves as a control-check for downstream reconciliation processes.
- Auditability: The system must be able to reproduce counts for a run by retaining the resolved SQL text/value for `\$\$M\_SRC\_SQL` and other relevant run identifiers.

Non-Functional Requirements

- Performance: The aggregation must complete within the batch SLA for expected source volumes.
- Reliability: If `\$\$M\_SRC\_SQL` fails to resolve or returns invalid SQL, the job must fail fast with clear error messaging.
- Maintainability: `\$\$M\_SRC\_SQL` must be version-controlled, and changes must be reviewed as they define the reconciliation population.

Open Items / Clarifications Required

1. Confirmation on whether the required count is `COUNT()` or `COUNT(T\_TX\_REQUEST\_REQUEST\_STATUS)`.
2. Specifications for the flat file target, including path, naming convention, delimiter, and overwrite vs. append behavior.
3. Validation of whether a precision of 10 is sufficient for `SOURCE\_RECT` based on expected maximum count sizes.
4. Exact SQL or logic generating `\$\$M\_SRC\_SQL`, including any filters such as process date, status, or source system.